Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

CHIANG RAI WORKSHOP · 20 kWp · PHASE-BALANCE REPORT

Why the inverter cannot use all its solar when the load is unbalanced

Solis S6-EH3P20K-H 20 kW three-phase high-voltage hybrid · zero export · data July–October 2026 · Rev 01 · 30 October 2026

80%
load imbalance on normal days (L1 58%, L2 12%, L3 30%)

28.9 A
datasheet maximum per phase. L1 peaked at 27.6–30 A

11.5 kW
the most the inverter can usefully deliver at 58% on L1, out of 20 kW

600–1,000
kWh/month of solar not produced under zero export

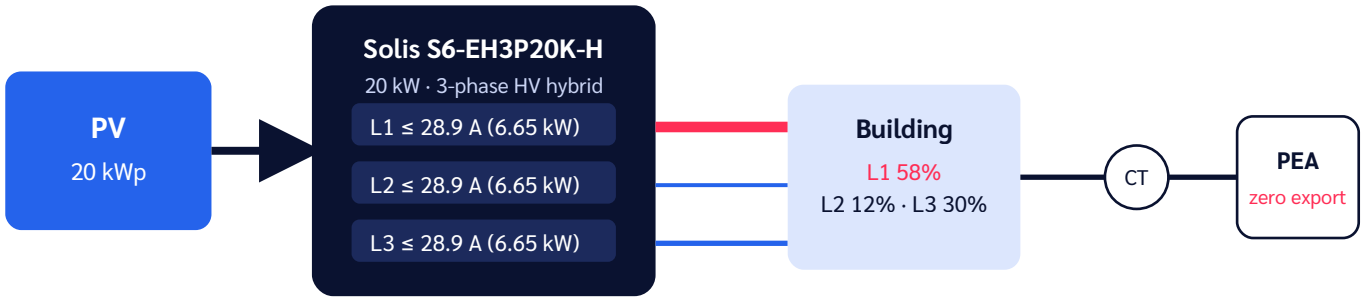
The short answer

One phase is full
Under zero export each phase may only receive what it uses, up to 28.9 A (about 6.65 kW). L1 carries most of the load and reaches that limit first.

The others are empty
L2 and L3 use very little, so the spare solar has nowhere to go and the inverter has to turn itself down.

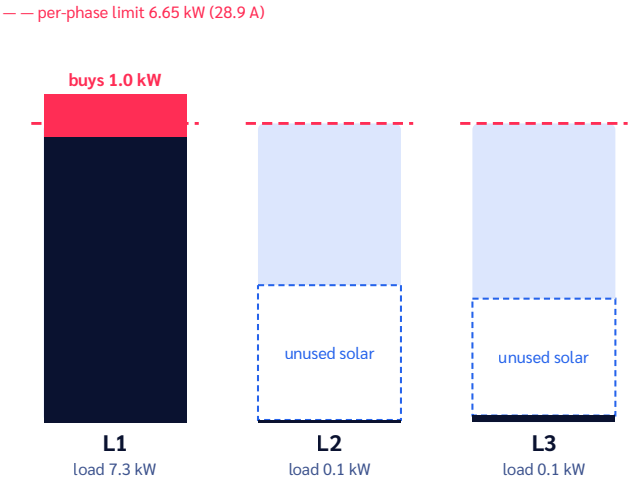
No fault, no trip
The datasheet allows 100% phase imbalance. The inverter keeps running, but it delivers far less than it could.

1 How the system is connected



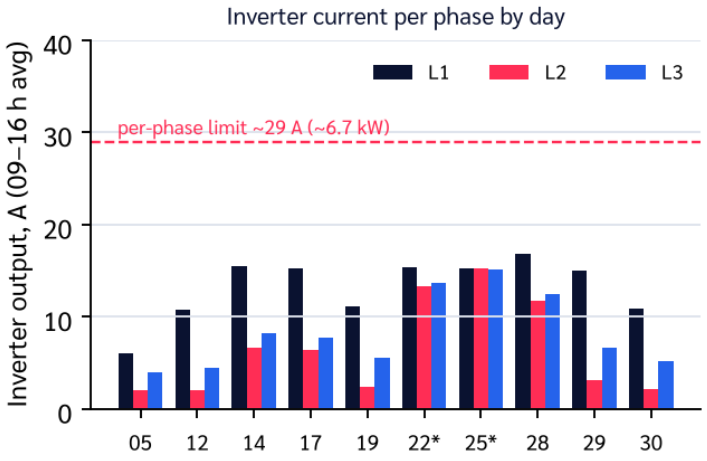
The panels feed one 20 kW inverter. The inverter sends power to each of the three phases separately, but no phase can take more than 28.9 A. A meter (CT) at the PEA connection tells the inverter how much each phase is buying, and zero export means the inverter must never push power back to PEA.

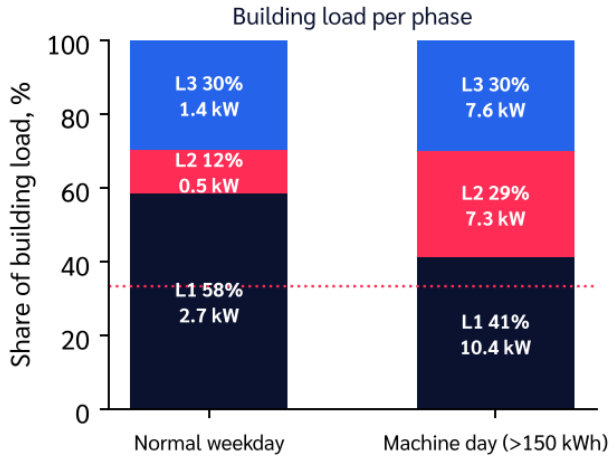
2 What happens on an unbalanced phase



This is a snapshot of 29 October, 13:00–14:20. L1 is at its limit (26–28 A) and still buys about 1 kW from PEA. L2 and L3 are almost idle at 0–0.6 A. The panels could give 11–14 kW (they did on 22–28 October), but the inverter delivers only about 6 kW.

> **The bottleneck is L1, not the panels and not the inverter as a whole.** The inverter has spare capacity on L2 and L3, but those phases have no load to use it, and zero export does not allow it to be sent to PEA.



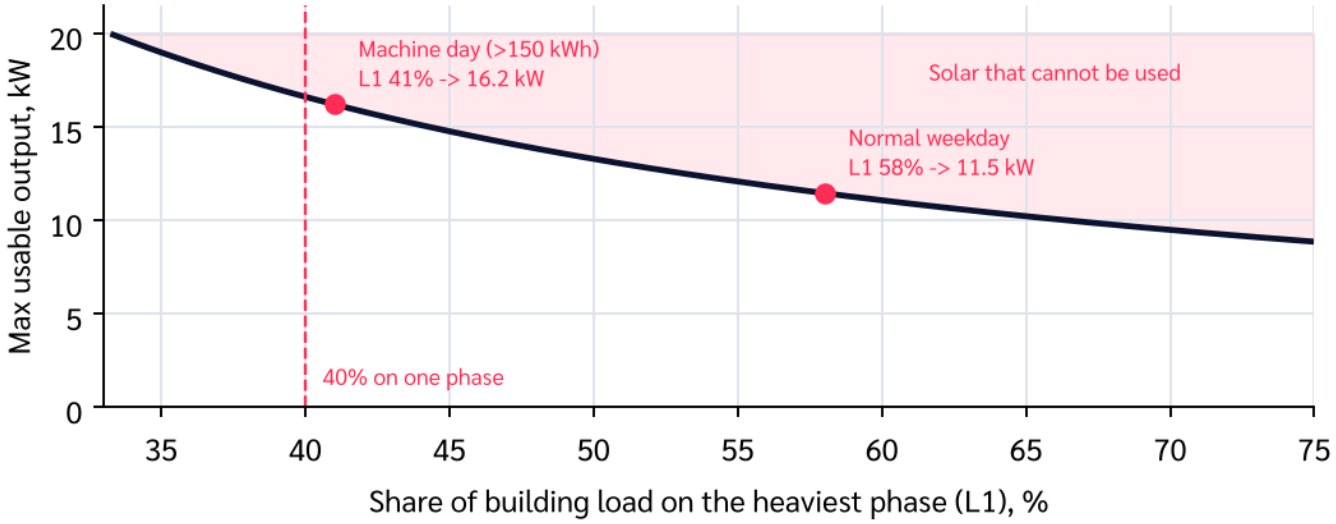


Left: average inverter current per phase; * = days when export was allowed, so output is equal on all phases. Right: average building load per phase; the dotted line is a perfectly balanced 33%.

3 What the datasheet says

20 kW	28.9 A	100%	40–50%
rated output, total across 3 phases	maximum output current per phase at 230/400 V, about 6.65 kW	maximum phase imbalance allowed, so no shutdown	maximum power on one phase, as % of rating

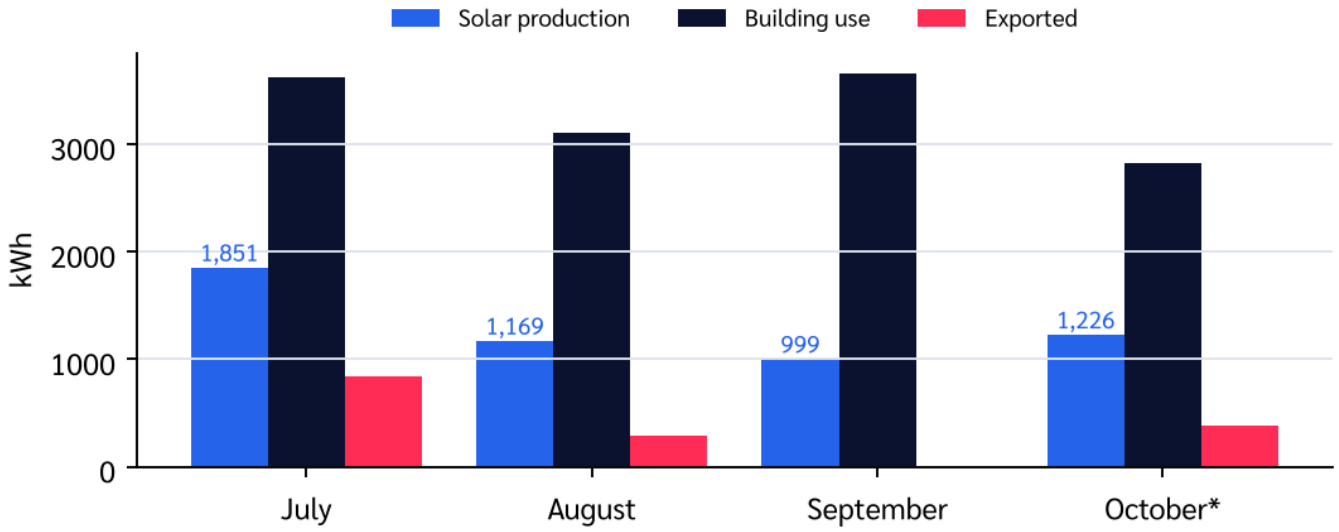
The most solar the building can use is the sum, across the three phases, of either that phase's load or 6.65 kW, whichever is smaller. Once the heaviest phase carries more than about 40% of the load, the usable output falls below 83% of the rating. At this site's 58% it falls to 11.5 kW, about 57%. The shaded area in the chart is solar the inverter cannot use.

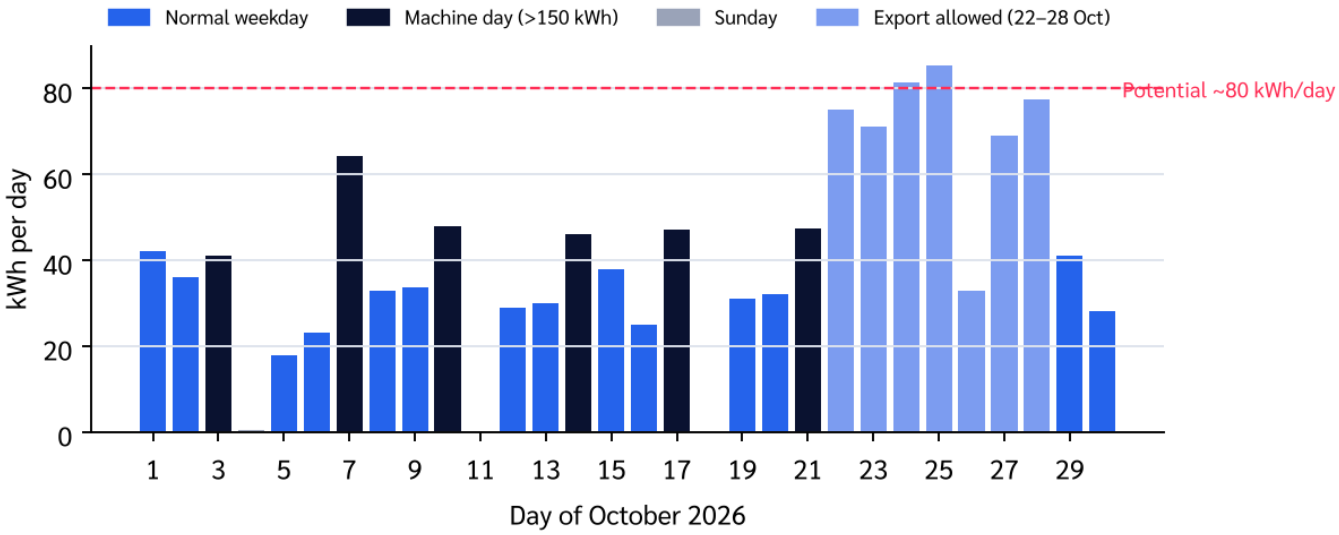


4 Common claims, checked against this inverter and this site

Claim	What the datasheet and the site data show
Over 40% imbalance makes the inverter shut down to protect itself.	**Not for this model.** The datasheet allows 100% phase imbalance on the grid and backup ports, and in October the inverter ran every day without a trip. It turns down; it does not trip.
A three-phase inverter splits its capacity evenly: a 12 kW unit gives only 4 kW per phase.	**Partly true.** The per-phase current limit here is 28.9 A, about 6.65 kW, which is one-third of 20 kW. That ceiling is what L1 hits. Below the ceiling, the inverter gives each phase a different amount.
You must enable "Unbalanced Output".	**Useful, and already working.** The grid reading on each phase stays near 0 W while the phase currents differ (L1 15 A, L2 3 A), which shows it is on. Confirm it at Grid Feed-in Power Limit – EPM.
Check that the CT clamps face the right way and sit on the right phases.	**Valid.** The main meter was wired to the wrong phases until 30 September, which distorted the September figures. It is correct now.
Replace it with a 100% unbalanced-output inverter, add a phase-balancing transformer, or use three single-phase inverters.	**Not needed here.** These cost far more. Rebalancing the loads on the board fixes the L1 limit at almost no cost.

5 What it costs in production





Lost production: compared with what the plant made when export was allowed, roughly **600–1,000 kWh/month** of solar is not produced under zero export. This is an estimate based on 30 days of data.

6 What to do, in order of value

1. **Rebalance the single-phase loads** No cost
   Move about 1–1.5 kW of daytime single-phase load (an air conditioner, a water heater or a machine circuit) from L1 to L2 at the distribution board, then measure each phase with a clamp meter.
2. **Confirm the inverter settings** No cost
   Unbalanced Output ON, export limit and meter/CT direction as recorded. Keep these settings in the site log.
3. **Run machines in sunny hours** No cost
   Run flexible machine work from 10:00 to 15:00, so the solar goes straight into the load.
4. **Add a high-voltage battery** Investment
   The inverter already has a 120–800 V battery port at up to 20 kW. A battery stores the solar the phases cannot use and supplies it in the evening.
5. **Apply to sell surplus to PEA** Application
   With export allowed, the output balances across the phases and reaches 11–14 kW, as it did on 22–28 October.

> **In one sentence:** the inverter is healthy and designed for unbalanced loads. It produces less because zero export plus a load concentrated on L1 leaves it with nowhere to send the solar. Balance the phases first, then store or export the surplus.

References

1. Solis — S6-EH3P(12-20)K-H datasheet, V2.2 (11/2024)
   https://solisinverters.com/uploads/file/Solis_datasheet_S6-EH3P(12-20)K-H_VNM_V2,2_202411.pdf
2. Solis — S6-EH3P(12-20)K-H datasheet, V1.8 (10/2024)
   https://solisinverters.com/uploads/file/Solis_datasheet_S6-EH3P(12-20)K-H_CHL_V1,8_202410.pdf
3. Solis — Zero export solutions: balancing three-phase loads (C&I)
   https://www.solisinverters.com/in/documentation/Maximizing_Efficiency_with_Solis_Zero_Export_Solutions_Balancing_Three-Phase_Loads_for_C_I_Applications_in.html
4. Solis — S5/S6 hybrid inverters: zero-export setting (Unbalanced Output)
   https://solis-service.solisinverters.com/en/support/solutions/articles/44002158700-solis-s5-s6-hybrid-inverters-zero-export-set
5. Hopewind — Zero output function under unbalanced load
   https://support.hopewind.com/portal/en/kb/articles/how-will-the-zero-output-function-respond-in-the-event-of-an-unbalanced-load
6. Fronius — Asymmetric production application guide
   https://fronius.com/~/downloads/Solar%20Energy/Application%20Guides/SE_APG_Asymmetric_Production_Fronius_Inverters_EN.pdf
7. SolisCloud — 20 kWp workshop plant, Chiang Rai, inverter and meter per-phase data, July–October 2026
   Monitoring data, downloaded 30 October 2026

Per-phase kW are estimated at 230 V without power factor. Figures are estimates from monitoring data, not a guarantee. Equipment and settings should be verified on site by a qualified electrician.